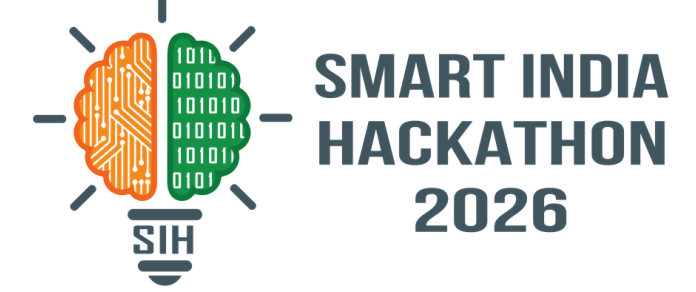


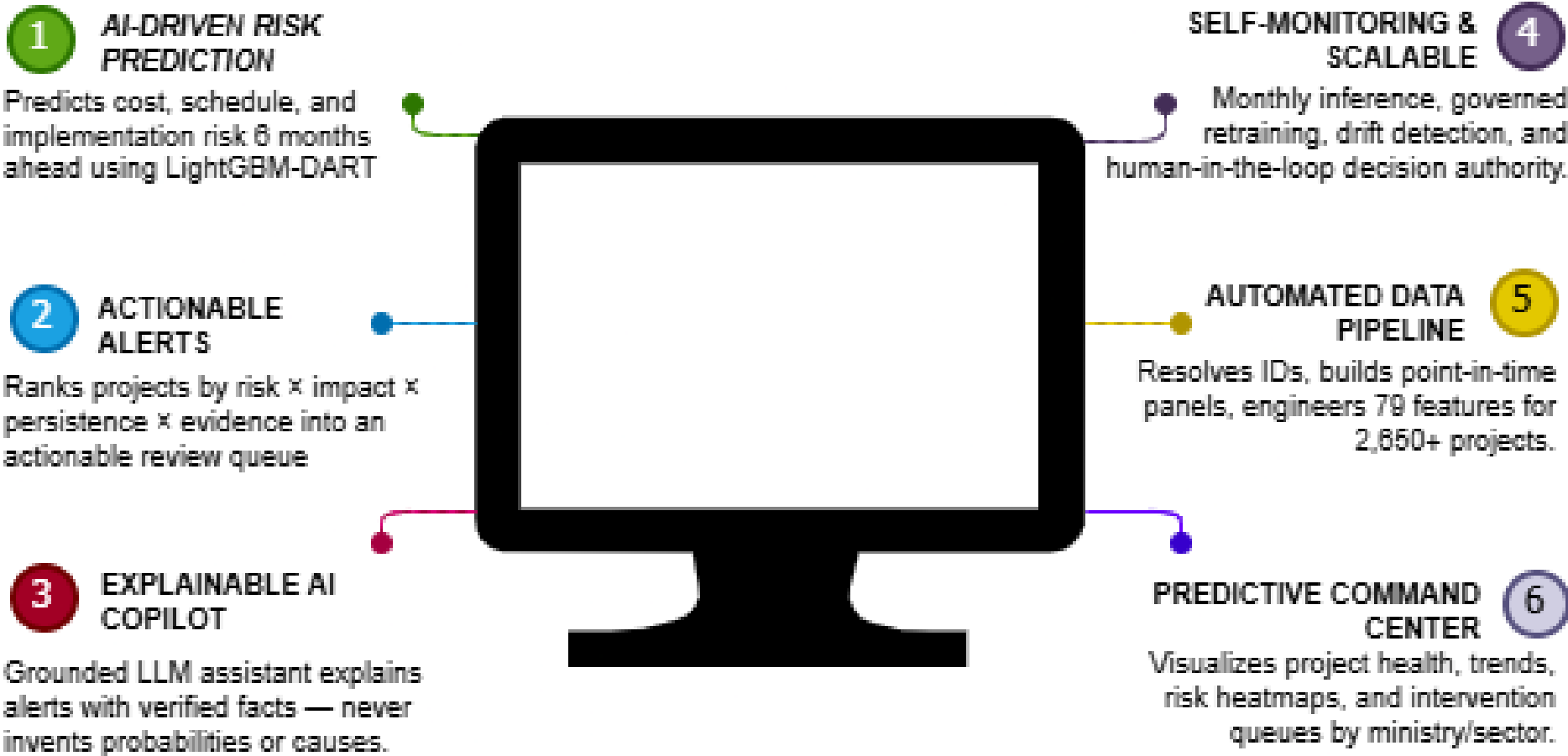
SMART INDIA HACKATHON 2026



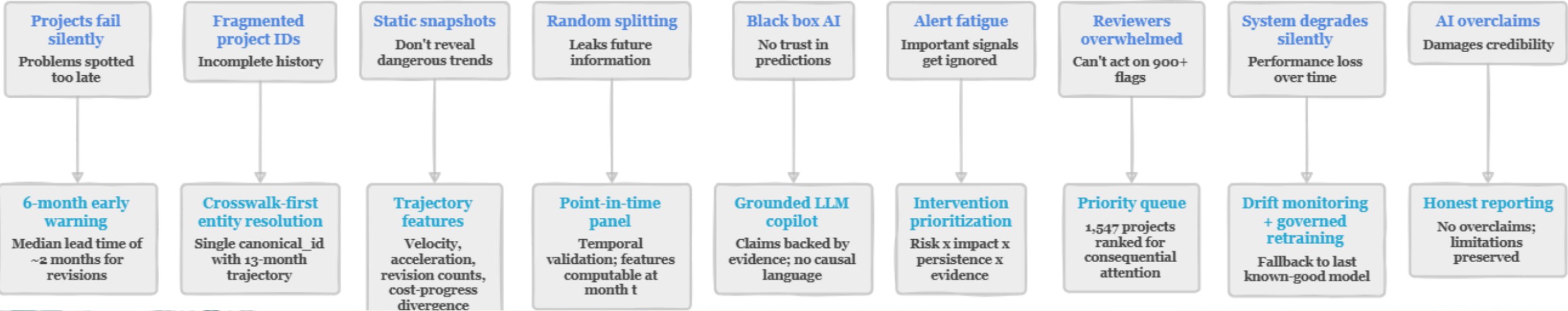
- Problem Statement ID –26103
- Problem Statement Title- [Use case on web-based integrated project monitoring](#)
- Theme- Smart Automation
- PS Category- Software
- Team ID- [125829](#)
- Team Name- [Code&Cook5](#)



PROPOSED SOLUTION



- Human Governance**
System recommends, officials decide
- Government Context**
Handles specific Indian challenges
- Forensic Reconciliation**
Audits competing model pipelines
- Point-in-Time Discipline**
Uses only available data
- Intervention Prioritization**
Focuses on high-impact risks
- LLM Copilot**
Explains predictions with evidence
- Trajectory Analysis**
Measures project behavior over time
- Proactive Planning**
Estimates cost and schedule



INNOVATION & UNIQUENESS

PROBLEMS ADDRESSED



TECH
STACK

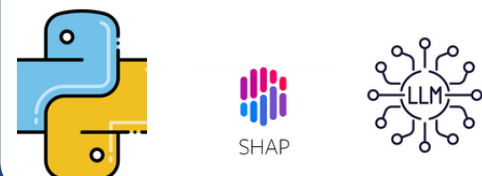
FRONTEND



Backend & API



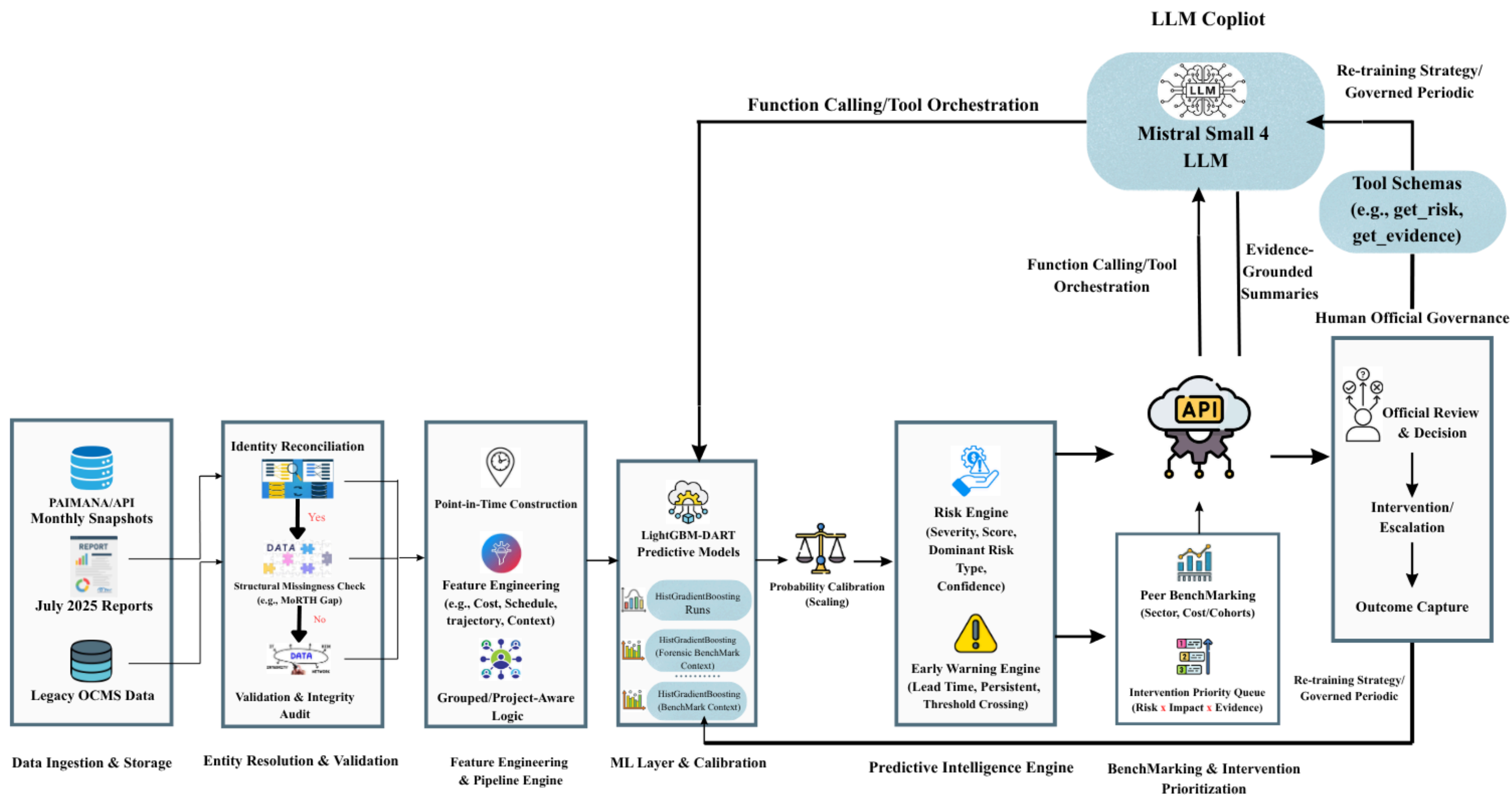
AI / ML



Data & Deployment



TECHNICAL ARCHITECTURE



FEASIBILITY AND VIABILITY

Predict Early • Explain Clearly • Act on Time



FEASIBILITY

Can we build and deploy it?

- **Technically Feasible:** Proven ML & analytics tools
- **Cost-Effective:** Built with open-source technologies
- **Easy Integration:** Connects with existing project data



VIABILITY

Will it work in the real world?

- **Target Users:** Government & project-monitoring agencies
- **Scalable:** Works across ministries, sectors & states
- **Continuous Learning:** Improves with new project data



CHALLENGES

What are the key hurdles?

- **Data Quality:** Missing or inconsistent project data
- **Model Accuracy:** Different sectors have different risk patterns
- **User Adoption:** Need simple, explainable AI insights



SOLUTIONS

How do we address them?

- **Data Preprocessing:** Clean and validate project data
- **Model Validation:** Compare and evaluate multiple ML models
- **Explainable AI:** Show the reasons behind predictions



USE CASES

Where can it make a difference?

- **Risk Prediction:** Predict cost & time overruns
- **Early Warnings:** Detect risk signals before escalation
- **Project Prioritization:** Focus attention on high-risk projects



IMPLEMENTATION POTENTIAL

Can it be adopted and scaled?

- **Fits Existing Systems:** Complements current monitoring workflows
- **Highly Scalable:** Expand across projects and departments
- **Continuous Improvement:** Refine predictions with new outcomes

Building
a Stronger
India Together

Monitoring Today
for a Better Tomorrow



Roads

Better Connectivity



Bridges

Stronger Structures



Urban Infrastructure

Growing Cities



Energy

Reliable Supply



Water Resources

Sustainable Development

Scalable across
sectors, states
and ministries



Multiple
Sectors



Pan-India
Scalability



Greater
Public Value

IMPACT



Government & Policymakers

- **Early risk visibility:** Identify projects needing attention
- **Smarter Decisions:** Allocate resources based on risk

Impact: More proactive and efficient governance



Implementing Agencies

- **Early Issue Detection:** Act before risks escalate
- **Focused Monitoring:** Prioritize high-risk projects

Impact: Smoother project execution



Citizens & Communities

- **Faster Delivery:** Reduce avoidable project delays
- **Better Infrastructure:** Improve public value

Impact: Better Infrastructure and public value



Monitoring Today for a
Better Tomorrow

BENEFITS



Economic Benefits

- **Lower Cost Overruns:** Detect financial risks early
- **Better Budget Use:** Protect public investment

Outcome: More value from every rupee invested



Operational Benefits

- **Predict Delays Early:** Enable timely intervention
- **Automated Risk Alerts:** Reduce manual monitoring

Outcome: From reactive to proactive monitoring



Governance Benefits

- **Explainable Insights:** Understand why projects are risky
- **Data-Driven Governance:** Support evidence-based decisions

Outcome: More value from every rupee invested

RESEARCH & REFERENCES

1. PAIMANA – Infrastructure & Project Monitoring, MoSPI

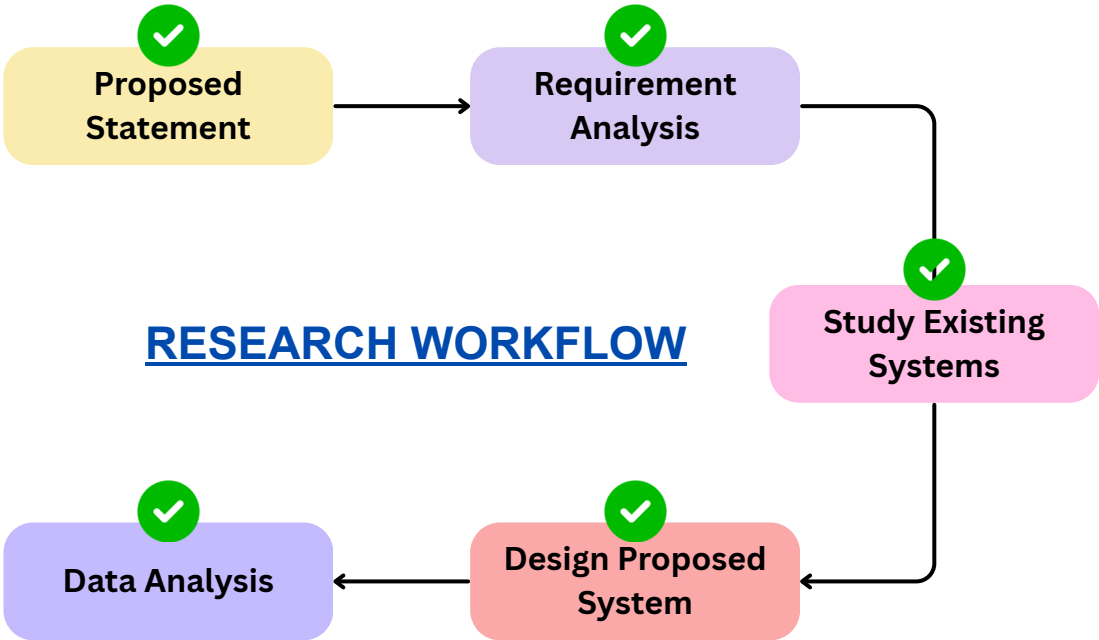
Official source for project monitoring data, project costs, expenditure, physical progress and completion information. [\[Link\]](#)
2. MoSPI – PAIMANA / OCMS Project Monitoring

Official Government of India project monitoring portal and dashboard. [\[Link\]](#)
3. Press Information Bureau – PAIMANA Web Portal

Reference for PAIMANA, OCMS modernization and API-based project data integration. [\[Link\]](#)
4. Scikit-learn – Machine Learning

Reference for Logistic Regression, HistGradientBoosting, Random Forest, Extra Trees and model evaluation. [\[Link\]](#)
5. SHAP – Explainable AI

Reference for SHAP-based global and local model explainability. [\[Link\]](#)



  **Live Demo:** <https://github.com/afreenAI/InfraSight-AI-Project-Monitoring-Platform>

Existing solutions vs PRAGATI
same data. smarter insights.

 Turning Project Data into Proactive Action

Capability	PAIMANA / OCMS (current MoSPI portal)	nPlan (private AI schedule-risk SaaS)	PRAGATI (Our Solution)
Cost / Progress Monitoring	✓	✗	✓
ML-based project risk scoring	✗	✓	✓
Cost overrun prediction	✗	Indirect	✓
Time overrun prediction	✗	✓	✓
Early-warning alerts	✗	✓	✓
Explanations (Why)	✗	Limited	✓
Calibrated confidence (not a raw score)	✗	Undisclosed	✓
Benchmarking	✗	✗	✓
Built on live PAIMANA / CUF data	✓	✗	✓
LLM copilot	✗	✓	✓
Access	Public, read-only	Paid enterprise	Open, free for govt use

